

8 (a) When a clean metal surface is illuminated by electromagnetic radiation, electrons may be emitted from the surface.

(i) State the name of this phenomenon.

..... [1]

(ii) State what is meant by the threshold frequency of the metal.

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..... [1]

(iii) Explain why the existence of the threshold frequency provides evidence for the photon model of electromagnetic radiation.

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..... [3]

(b) When a clean metal sheet is illuminated by radiation of wavelength 254 nm, electrons are emitted with a maximum kinetic energy of 1.91×10^{-19} J.

When the same metal sheet is illuminated by radiation of wavelength 178 nm, the maximum kinetic energy of the emitted electrons is 5.27×10^{-19} J.

(i) Use these data to calculate a value, to three significant figures, for the Planck constant.

Planck constant = Js [3]

- (ii) Determine a value, in eV, for the work function energy of the metal. Give your answer to two significant figures.

work function energy = eV [3]

[Total: 11]